



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2025-26(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Data Warehousing and Mining

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33505	Course Outcome
CO1	To classify the fundamentals, need, structure, and cost of data warehousing for effective data management.
CO2	To classify data warehousing architecture, modeling techniques, ETL processes, and OLAP operations for efficient data analysis.
CO3	To explain data preprocessing techniques and transformation for improving data quality.
CO4	To provide fundamental knowledge of data mining concepts, techniques, and applications.
CO5	To introduce classification techniques and equip students with the ability to compare and apply different classifiers effectively.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define OLAP.	CO1	2	2
b.	What is fact table?	CO2	2	2
c.	Define data transformation.	CO3	3	2
d.	What is association rule mining?	CO4	2	2
e.	Define Big Data analytics.	CO5	2	2

Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain structure and architecture of data warehouse.	CO1	2	5
b.	Discuss cost factors involved in data warehousing.	CO1	2	5
c.	Explain role of data warehouse in decision support systems.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain ETL process with challenges.	CO2	2	5
b.	Compare star, snowflake, and fact constellation schema.	CO2	3	5
c.	Explain OLAP operations (roll-up, drill-down, slice, dice).	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain data cleaning process in detail.	CO3	4	5
b.	Discuss data reduction techniques	CO3	3	5
c.	Explain preprocessing pipeline with example.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain types of patterns mined in data mining.	CO4	2	5
b.	Discuss major issues in data mining.	CO4	4	5
c.	Explain applications of data mining in real-world scenarios.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain BI framework and development process.	CO5	2	5
b.	Discuss Hadoop architecture and working.	CO5	3	5
c.	Explain Big Data vs traditional data systems.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create